

T LALITH KRISHNA

School of Life Sciences, Mahindra University, Bahadurpally, Hyderabad, India – 500043
Mobile: +91 9550770444 | lalithkrishna747@gmail.com | se23cab012@mahindrauniversity.edu.in
linkedin.com/in/t-lalith-krishna-7a189a334

EDUCATION

B.Tech in Computational Biology — Mahindra University, School of Life Sciences, Hyderabad *Aug 2023 – Present*
Intermediate (Class XII) — Aakash Institute, Hyderabad *June 2021 – 2023*
Higher Secondary (Class X) — St. Paul's High School, Hyderabad *March 2021*

LEADERSHIP & PROJECT DELIVERY

Founder — ClarityFi | Personal Finance & Mindfulness Platform *2024 – Present*

- Owned the full product lifecycle solo — planning, scoping, build, and rollout — of a full-stack web application combining budget tracking, goal management, and mindfulness tools.
- Grew the user base to 50+ active users organically through iterative releases based on usage feedback.

Outreach Head — ISCB Student Chapter *2025 – Present*

- Coordinate partnership outreach and stakeholder engagement across the International Society for Computational Biology chapter at Mahindra University, aligning communications with chapter goals and timelines.

Co-Founder — Tiny Triumphs Childcare | Early-Stage Childcare Venture *2024*

- Conceptualised and launched a childcare service platform end-to-end including a custom booking system from initial planning through go-live.

RESEARCH & TECHNICAL PROJECTS — PROCESS & DELIVERY HIGHLIGHTS

- ToxINsight:** Full-stack web server and relational database for snake venom proteins, integrating sequence, structural, functional, interaction, and computational analysis data.
- SVMP Antivenom Computational Study:** Identifying peptide inhibitors against snake venom metalloproteinases (SVMPs) using molecular docking, structural analysis, and protein–peptide interaction studies.
- ML Cell Classification:** Developing a supervised machine learning pipeline using Cellpose and EfficientNetV2-S to classify senescent versus non-senescent cells from microscopy images.
- Miltefosine Resistance Genomics:** Investigating drug resistance mechanisms in *Leishmania donovani* using evolutionary genomics, comparative analysis, and machine learning.
- OSCC RNA-Seq Splicing Analysis:** Identifying differentially expressed RNA splicing factors in oral squamous cell carcinoma (OSCC) through RNA-Seq data analysis.
- Diabetes Genetics Relational Database:** Designing a normalized relational database linking diabetes types with associated genes, inheritance patterns, and related diseases.

- **Leish-Amplify:** Building a Snakemake-based HPC/SLURM pipeline to detect extrachromosomal circular DNA (eccDNA) in drug-resistant *Leishmania donovani* strains, including debugging and optimization of the filtering workflow.

SKILLS

Project & Delivery: End-to-end product ownership, scoping and planning, stakeholder communication, process auditing/QA, documentation.

Technical: Python, R, SQL, Linux/Bash, Snakemake/HPC pipelines.

Other: Market research, full-stack web development, AI tools for research and workflow automation.

ACHIEVEMENTS

- National-level roller hockey player.